

Q.1.

1. What is primary key?
2. What is Foreign Key and why it is needed?



1. What is primary key?



A **primary key** is a specific type of database constraint used in relational databases to uniquely identify each record within a table. Here are the key characteristics and features of a primary key:

Key Characteristics:

1. **Uniqueness:** Each value in the primary key column (or columns) must be unique across the table, meaning no two records can have the same primary key value. This ensures that each record can be uniquely identified.
2. **Non-null:** A primary key cannot contain null values. Every record in the table must have a valid primary key value.

3. **Immutable:** The value of a primary key should not change. Once assigned, it is best practice to keep the primary key value constant throughout the life of the record.

4. **Single or Composite:** A primary key can consist of a single column (single primary key) or multiple columns (composite primary key). A composite primary key is used when a single column cannot uniquely identify a record.

2.What is Foreign Key and why it is needed?



A **foreign key** is a column or a set of columns in a relational database table that establishes a link between data in two tables. It is a key used to connect the data in one table to the data in another table, thereby enforcing referential integrity between the two tables.

Key Characteristics of Foreign Key:

1. **Reference to Primary Key:** A foreign key in one table refers to the primary key of another table. This creates a relationship between the two tables.
2. **Allows Duplicates and Nulls:** Unlike primary keys, foreign keys can have duplicate values and can also accept null values (unless explicitly constrained). This allows for relationships where one record in the primary table can relate to multiple records in the foreign key table.
3. **Data Integrity:** Foreign keys enforce referential integrity by ensuring that the value in the foreign key column matches an existing value in the primary key column of the referenced table.

Why Foreign Keys Are Needed:

1. **Data Integrity:** Foreign keys ensure that relationships between tables remain consistent. If a foreign key value does not correspond to any primary key in the referenced table, it indicates a data integrity violation.

2. **Establish Relationships:** Foreign keys allow for the creation of relationships between different tables, which is fundamental in relational database design. They enable users to perform complex queries that involve multiple tables.
3. **Prevent Orphan Records:** By enforcing referential integrity, foreign keys help prevent orphan records in the child table (i.e., records that reference non-existent records in the parent table).
4. **Cascading Actions:** Foreign keys can be set up with cascading actions (like ON DELETE CASCADE or ON UPDATE CASCADE), which automatically update or delete related records in the child table when changes occur in the parent table.